Title: Case Study: Academic Stress Analysis

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Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn), Tableau.

🎯 Objective

To analyze the relationship between academic stress levels and key influencing factors such as peer pressure, academic pressure from home, study environment, and coping strategies among students.

📊 Key Findings (KPIs)

- **Average Stress Index:** 3.7 → indicates relatively high stress levels.
- **Maximum Stress Index:** 5 → some students experience extreme stress.
- **Peaceful Environment %:** 49.6% → less than half of student's study in calm environments.

🔍 Detailed Insights

- **Peer Pressure vs Stress Index**
  Stress levels rise consistently with higher peer pressure, showing peer influence as a significant contributor.
- **Academic Pressure vs Stress Index**
  Academic pressure from home has a stronger correlation with stress than peer pressure, highlighting family expectations as a key driver.
- **Study Environment vs Stress Index**
  Students in peaceful environments report lower stress. Noisy or disrupted settings are linked to higher stress and greater variability.
- **Coping Strategies**

- 63% rely on rational coping (analyzing and handling situations intellectually).
- 22% experience emotional breakdowns.
- 15% depend on social support (friends, family).
  This suggests a gap in external support networks.

📈 Implications

- **Interventions** should focus on improving study environments and balancing academic expectations.
- **Support Programs** are needed to encourage healthier coping mechanisms and expand social support.
- **Policy Recommendations**: Schools and universities should integrate stress management workshops and provide counseling services.

📝 Conclusion

This case study demonstrates that academic stress is influenced by multiple factors, with academic pressure and study environment playing the most critical roles. Addressing these areas can significantly reduce stress levels and improve student well-being.